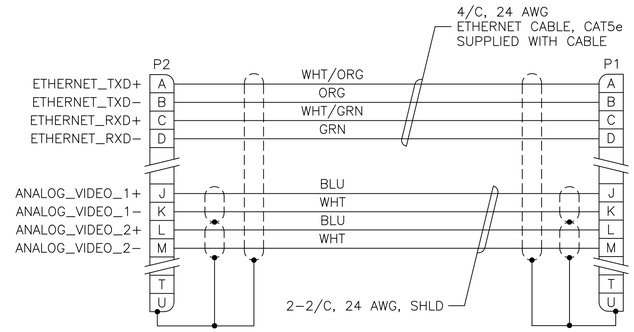
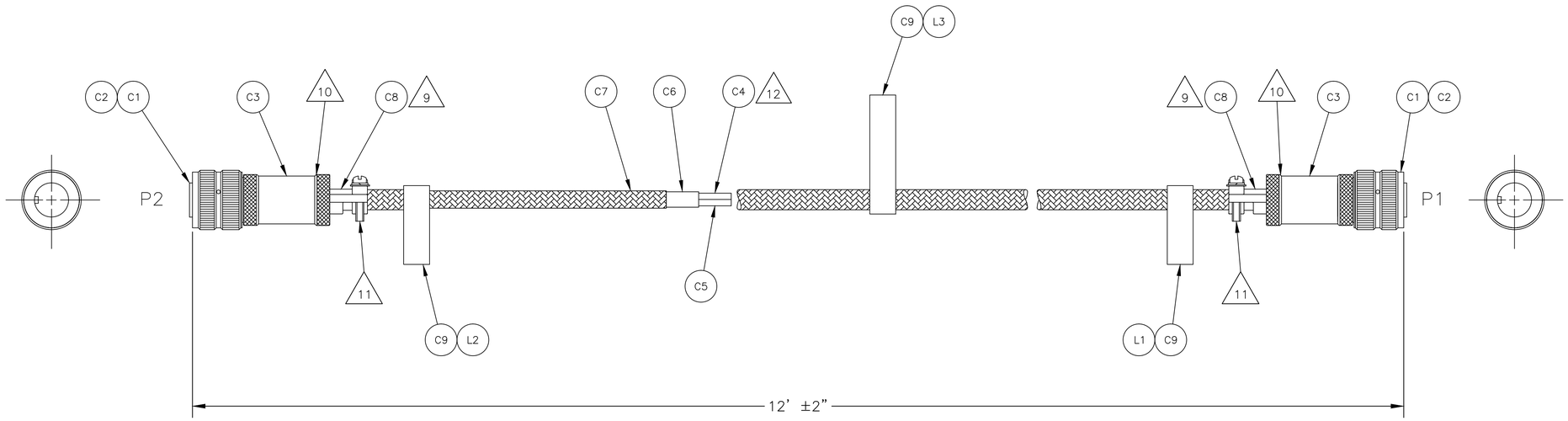
COMMERCIAL PARTS LIST

ITEM	QTY	PART NUMBER	DESCRIPTION	MANUFACTURER
C1	2	D38999/26FD18SN	CONNECTOR, CIRCULAR PLUG, 18 POS.	QPL
C2	A/R	M39029/56-351	CONTACT, SOCKET, SIZE 20, AWG 20-24	QPL
C3	2	M85049/1915N05	BACKSHELL, STRAIGHT, SIZE 15	AMPHENOL
C4	1	73-7793-14	CABLE, ETHERNET PATCH CAT5e, 14'	EMERSON
C5	A/R	M17/176-00002	2/C, 24 AWG, TWINAXIAL	QPL
C6	A/R	5011001332S	SLEEVING, EXPANDABLE BRAID, 1/2" DIA.	BENTLEY HARRIS
C7	A/R	2146	TINNED COPPER BRAID, 1/2"	ALPHA
C8	A/R	M23053/13-004-0	TUBING, VITON, 1/2" DIA.	QML
C9	A/R	TZE-211	LABEL, P-TOUCH, 1/4", BLACK ON WHITE	BROTHER
C10	A/R	2216 B/A GRAY	EPOXY, 2 PART	3M
C11	A/R	7649	PRIMER N	LOCTITE
C12	A/R	222MS	THREADLOCKER	LOCTITE

LABEL LIST	
ITEM	INSCRIPTION
L1	P1
L2	P2
L3	S1101EGA4558 SN xxxx



SCHEMATIC



NOTES:

1. MATERIAL CERTIFICATION REQUIRED PER QUALITY LEVEL SPECIFIED IN WORK ORDER/SOW.
2. ESD HANDLING AND CONTROL PER ESTABLISHED PROCEDURES.
3. HANDLE ALL PARTS AND ASSEMBLIES WITH CLEAN, LINT-FREE, ANTI-STATIC GLOVES OR FINGER COTS.
4. HARNESS ASSEMBLY AND INSPECTION SHALL BE IN ACCORDANCE WITH NASA-STD-8739.4 AND NASA-STD-8739.3 AND FOLLOWING NOTES:
   a. TO PREVENT POTENTIAL DAMAGE (COLD FLOW) OF THE UNDERLYING WIRE INSULATION, SEPARATOR SHALL BE APPLIED OVER THE WIRE BUNDLE TO GIVE THE WIRE CONTINUOUS PROTECTION.
   b. ALL SHIELD TERMINATIONS ARE INTERFERENCE SENSITIVE.
5. FOR TENSILE PULLS SEE TENSILE PULL STRENGTH LIST. TENSILE PULL STRENGTHS PER NASA-STD-8739.4 AND/OR THE MANUFACTURER'S SPECIFICATIONS.
6. INSTALL CONTACTS IN ALL UNUSED POSITIONS OF CIRCULAR CONNECTORS.
7. PERFORM CONTACT RETENTION TEST PER NASA-STD-8739.4 AND/OR THE MANUFACTURER'S SPECIFICATIONS. CONTACTS REQUIRING A PULL TEST FOR RETENTION SHALL BE PERFORMED AS FOLLOWS: A GENTLE PULL ON THE WIRE SHALL BE USED TO DETERMINE IF THE CONTACT IS SEATED PROPERLY.
8. TERMINATE SHIELDS INTO BACKSHELL PER MANUFACTURER'S SPECIFICATION.
9. TO SAFELY SECURE CABLE IN STRAIN RELIEF INCREASE THE CABLE DIAMETER USING HEAT SHRINK TUBING (ITEM C8). MAXIMUM OF TWO (2) LAYERS.
10. TORQUE BACKSHELL (ITEM C3) TO 40 ±5 IN-LBS. STAKE BACKSHELL USING EPOXY (ITEM C10).
11. TRIM FASTENERS ON STRAIN RELIEF TO EXPOSE 1-2 THREADS. APPLY PRIMER N (ITEM C11) AND THREADLOCKER 222MS (ITEM C12), TORQUE FASTENERS PER CHART. APPLY EPOXY (ITEM C10) TO EXPOSED THREADS.
12. REMOVE CONNECTOR FROM BOTH ENDS OF ITEM C4 AND DISCARD FOR CONNECTION OF ITEM C1.
13. TEST PER NASA-STD-8739.4 USING THE FOLLOWING LEVELS:
    CONTINUITY: < 5 OHMS
    DWV: VOLTAGE: 1500VDC ±75VDC
    CURRENT: < 1 MILLIAMP
    DURATION: < 1 MINUTE
    IR: VOLTAGE: 500VDC ±50VDC
    RESISTANCE: > 100 MEG OHMS
    DURATION: < 1 MINUTE
14. PLACE COMPLETED ASSEMBLY IN CLEAN, STATIC-SHIELDING PACKAGING. IDENTIFY PART BY TAGGING WITH PART NUMBER AND SERIAL/LOT NUMBER AS SPECIFIED IN WORK ORDER PER SAE AS478, METHOD 35D. STORE IN A PROJECT APPROVED STORAGE AREA.

CRIMP TOOL LIST				
CONTACT P/N	CRIMP TOOL	POSITIONER	I/E TOOL	REF. DES.
M39029/56-351	M22520/1-01 OR M22520/2-01	M22520/1-04 OR M22520/2-10	M81969/14-02	P1, P2

TENSILE PULL STRENGTH LIST			
MANUFACTURER	CONTACT P/N	WIRE AWG	LBS
QPL	M39029/56-351	24	8

BACKSHELL CLAMPING TORQUE TABLE		
SCREW SIZE	TORQUE RANGE	UNITS
2-56	2 ±0.5	IN-LB
4-40	4 ±0.5	IN-LB
6-32	6 ±1.0	IN-LB
8-32	8 ±1.0	IN-LB
10-32	10 ±1.0	IN-LB
.250-20	12 ±1.0	IN-LB

NEXT ASSEMBLY

UNLESS OTHERWISE SPECIFIED:
1. ALL DIMENSIONS ARE IN INCHES
2. BREAK EDGES .010-.020
3. INSIDE CORNERS R.02 MAX
4. LINEAR TOLERANCES:
   X ± 0.03
   .XX ± 0.01
   .XXX ± 0.005
5. ANGULAR TOLERANCE ± 0.5°

THIRD ANGLE PROJECTION

THIS DRAWING IS UNDER CONFIGURATION CONTROL; DO NOT REVISE WITHOUT AN APPROVED ECO.
APPROVED: E. McCLOSKEY 9/24/15
PRIOR TO USE, CHECK WITH CONFIGURATION MANAGEMENT FOR ANY OUTSTANDING ECOS.

DESIGNER	M. HOMKO 7/22/14	QUALITY	D. SEELEY 7/29/14
REVIEWER	J. BRUEWER 7/23/14	MATERIALS	M. WILKINSON 7/23/14
MECHANICAL	A. DRAKE 7/23/14	PROJ LEAD	B. BOROWSKI 7/29/14
ELECTRICAL	T. GOBELI 7/24/14	ENGINEER	M. SAVINA 7/22/14
STRUCTURAL	C. WERNER 7/23/14		
SAFETY	W. BORRELLI 7/22/14		

DIMENSIONS AND TOLERANCING PER ASME Y14.5M-1994

ZIN TECHNOLOGIES
6745 ENGLE ROAD
AIRPORT EXECUTIVE PARK
CLEVELAND, OHIO 44130
(440) 625-2200
www.zin-tech.com

CONTRACT NUMBER: NNC14CA02C
PROJECT CODE: S2-BT3

REV	DESCRIPTION	APPROVED
A	REVISED PER ACME-ECO-368.	MH 9/22/15 DS 9/22/15 TG 9/24/15

REVISION HISTORY

NATIONAL AERONAUTICS AND SPACE ADMINISTRATION
JOHN H. GLENN RESEARCH CENTER
LEWIS FIELD
CLEVELAND, OHIO

TITLE
OPTICS BENCH SIMULATOR ETHERNET/VIDEO HARNESS TO GSE TEST RACK

SIZE	CAGE CODE	DWG NO	REV
D	0SG68	S1101EGA4558	A

SCALE NONE | INITIAL REL DATE 07/30/14 | SHEET 1 OF 1

RELEASE LEVEL: RELEASED
MODEL NAME: S1101EGA4558_REV_A.DWG

AutoCAD 2012

DWG NO S1101EGA4558 | SH 1 | REV A